ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

Version 3.0 · February 2026*

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Competitive Benchmarking

1. MARKET OVERVIEW

Market Size & Growth

- Global Green Hydrogen Market: Rapid expansion, potentially reaching hundreds of gigawatts of capacity by the mid-2030s, fueled by climate policies and corporate net-zero commitments
- Current market size: **$3.8B (2023)**, projected to reach **$11.2B by 2027** (CAGR 30.7%)
- Electrolyzer capacity: **1.2 GW installed globally in 2023**, expected to reach **150+ GW by 2030**

Growth Drivers

1. Strong national policies and strategies (EU, U.S., China, Japan, Australia)
2. Corporate decarbonization goals
3. Decreasing costs of renewables (solar, wind)
4. Demand in hard-to-abate sectors (steel, ammonia, heavy transport)

End-Use Sectors

1. **Industrial**: Ammonia, refining, steel, chemicals (current dominant use)
2. **Transportation**: Fuel cell vehicles (buses, trucks, trains), maritime applications (growing segment)
3. **Energy Storage & Grid Balancing**: Long-duration storage from surplus renewable electricity (emerging opportunity)

Demand & Pricing Trends

- Falling electrolyzer CAPEX (<$500/kW) and rising efficiencies pushing green hydrogen toward cost parity with fossil-based hydrogen
- Current delivered hydrogen price: **$18–25/kg** (varies by region)
- Projected green hydrogen cost: **$1.50–3.00/kg by 2030** in optimal locations

2. REGULATORY & POLICY ENVIRONMENT

Government Incentives

- Tax credits, grants, loan guarantees for electrolyzer manufacturing and projects
- U.S. Inflation Reduction Act: **$3/kg production tax credit** for clean hydrogen
- Oklahoma Hydrogen Hub: **$150M in federal funding** for regional development

Standards & Certification

- Emerging standards for "green" or "renewable" hydrogen guide investor and purchaser decisions
- Oklahoma HB 1293: Streamlined permitting for hydrogen infrastructure

Oklahoma Context

- Abundant wind energy resources (**48.6% of state generation in 2023**)
- Potential for hydrogen infrastructure development
- Favorable business climate and skilled energy workforce

3. TECHNOLOGY LANDSCAPE

Established Electrolyzer Technologies

TECHNOLOGY	TYPICAL CAPEX ($/KW)	ELECTRICAL EFFICIENCY	MATURITY (TRL)	NOTABLE CHARACTERISTICS
Alkaline	$800–$900	60-65%	High (9)	Well-established, simpler design, moderate efficiency
PEM	$1,100–$1,300	55-65%	High (8-9)	Flexible, fast response, but higher cost & sensitive materials
Solid Oxide (SOEC)	$4,000+	70-80%	Mid (6-7)	High efficiency, high temp, complex, currently costly
AEM	>$900	55-60%	Mid (6-8)	Emerging, potential cost reductions, stability issues

Tobe Energy's Membrane-Free Electrolyzer Approach

METRIC	TOBE ENERGY ELECTROLYZER
CAPEX (Prototype)	**$446/kW**
Projected CAPEX (MW Scale)	**~$177/kW**
Stack Efficiency (LHV)	**94.7%**
System Efficiency (HHV)	**>92%**
System Energy Consumption	**~42 kWh/kg H_2**
Stack Lifetime	**~80,000+ hours** (est.)
Materials	304 Stainless Steel

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Key Differentiators:

1. **Stack efficiency of 94.7%** (LHV) and **>92% system efficiency** (HHV), exceeding current industry standards while maintaining practical operating conditions
2. **Low CAPEX**, projected to drop further with scale due to simplified design
3. **Durable, simple construction with no exotic materials**, lowering OPEX and downtime
4. **Membrane-free design** eliminates degradation issues common in PEM and AEM systems

4. COMPETITIVE LANDSCAPE

Leading Players & Strategies

- **Alkaline Leaders** (e.g., Nel, Thyssenkrupp): Focus on large-scale deployment and cost reduction through volume
- **PEM Innovators** (e.g., Plug Power, Cummins): Target flexible, rapid-response applications and integrated solutions
- **SOEC Pioneers** (e.g., Sunfire, Bloom): High efficiency as a differentiator, but need cost reductions and improved durability
- **AEM Startups** (e.g., Enapter): Emphasize modular systems with future cost reduction potential

Positioning Tobe Energy

METRIC	TOBE ENERGY	ALKALINE	PEM	SOEC	AEM
CAPEX	**Advantage**	Moderate	Weak	Very Weak	Moderate
Electrical Efficiency	**Outstanding** (94.7% stack / >92% system)	Low (~62%)	Low (~61.5%)	High (~78%)	Low (~57%)
Stack Lifetime & Costs	**Excellent**	Good	Moderate	Limited	Evolving
Scalability (Cost Reductions)	**Strong**	Good at scale	Improving	Uncertain	Promising but uncertain
Technology Readiness	Early but promising	Mature	Mature	Emerging	Emerging

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Strategic Insight: Tobe Energy can outperform incumbents with a breakthrough in both cost and efficiency, offering a next-generation solution rather than incremental improvements. The membrane-free design eliminates the single largest failure point in traditional electrolyzers while achieving superior efficiency metrics.

5. MARKET OPPORTUNITIES & STRATEGIC CONSIDERATIONS

Industrial Partnerships

- Immediate opportunities in ammonia, refining, steel sectors seeking green hydrogen
- Oklahoma industrial hydrogen demand: ~1,500 kg/day with 8% annual growth

Regional Advantages (Oklahoma)

- Integration with abundant renewables and supportive policies
- Oklahoma wind capacity: **11.4 GW** (4th in U.S.)
- Existing energy infrastructure and skilled workforce

Early Mover Advantage

- Differentiation through truly disruptive efficiency and cost metrics
- Only 3 electrolyzer manufacturers currently operating in Oklahoma

Scaling & Validation

- Demonstration projects at pilot and commercial scale to confirm CAPEX targets and reliability
- Tobe's **Q4 2026** deployment aligns with Oklahoma Hydrogen Hub timeline

6. USE CASE: GREEN HYDROGEN FOR AI DATA CENTERS

Green Hydrogen: The Key to Cost-Efficient, Zero-Emission AI Power

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Addressing AI's Growing Energy Needs: AI data centers are consuming energy at an unprecedented rate. As global warming concerns intensify, the pressure to adopt zero-emission, cost-effective energy sources is mounting. Nuclear, solar, and wind are prime contenders, but their variability and the need for large-scale energy storage solutions remain challenges.

Nuclear Energy & Off-Peak Pricing:

- Nuclear plants run consistently and don't easily adjust to fluctuating demand
- Utilities like Duke Energy offer steep off-peak discounts — over 60% cheaper than peak rates
- Cost-effective, large-scale energy storage remains a key hurdle

Hydrogen Storage vs. Batteries:

1. Traditional batteries are costly and rely on rare earth minerals
2. The Tobe Electrolyzer produces hydrogen at a fraction of battery storage costs — less than **$2,500** to store energy equivalent to what would require **$30,000** worth of batteries
3. Hydrogen acts like an energy "battery" but avoids the material constraints of lithium-ion storage

Why It Hasn't Been Done Before: Prior hydrogen technologies were inefficient and complex. The Tobe Electrolyzer is the next-generation solution, using novel power electronics and a reimagined design to slash costs and energy waste.

Impact on AI Data Centers

Cost Savings:

- Store energy as green hydrogen during off-peak hours when it's ~60% cheaper, then use it during peak periods to drastically cut electricity bills
- Typical AI data center energy cost reduction potential: **$1.2M/year per 10MW facility**

High Efficiency at Low Cost:

- Tobe's system is 75% cheaper to build and significantly more efficient than traditional electrolyzers, providing an affordable and sustainable solution for AI's massive energy demands

Zero-Emission Resilience:

- Utilizing nuclear power plus green hydrogen creates a zero-emission power cycle
- As hydrogen is converted back to electricity, the only by-product is water
- This reduces carbon footprints and eases grid stress by evenly distributing energy usage

Future-Proofing AI Expansion:

1. Capitalize on low-cost energy during off-peak hours
2. Eliminate reliance on costly battery systems and finite rare earth resources
3. Maintain a sustainable, flexible, and scalable energy model aligned with carbon reduction goals

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By integrating green hydrogen from Tobe Electrolyzers, AI data centers can address escalating power demands without compromising cost or sustainability. This synergy paves the way for a more stable, greener, and economically viable energy landscape for AI's future.

7. CONCLUSION

The green hydrogen market is rapidly evolving, with mounting pressure to lower costs and increase efficiency. Tobe Energy's membrane-free electrolysis stands out by combining unprecedented electrical efficiency (**94.7% stack** (LHV), **>92% system** (HHV)), significantly lower CAPEX (**~$177/kW** at scale), and robust materials. This positions Tobe Energy to capture early market share, attract investors, and shape the industry's future trajectory.

The AI data center use case highlights the versatility of green hydrogen as a cost-effective, zero-emission energy storage solution. By pairing nuclear off-peak electricity with Tobe's cutting-edge electrolysis technology, AI data centers — and other energy-intensive industries — can secure reliable, scalable, and environmentally responsible energy solutions.

In essence, the future of green hydrogen is Tobe:

1. **Disrupting the cost-efficiency barrier** with **94.7% stack efficiency** and low CAPEX of **~$177/kW**
2. **Providing a clear pathway to zero-emission energy storage** through practical deployment
3. **Empowering new markets like AI data centers to thrive sustainably** with innovative applications

- **Market Position**: Only membrane-free electrolyzer technology at commercial scale
- **Technology Edge**: 94.7% stack efficiency (LHV) vs. industry average of 60-78%
- **Financial Advantage**: **$1.81/kg LCOH** vs. market average of $4.50+/kg
- **Growth Potential**: $11.2B U.S. hydrogen market by 2027 with 30.7% CAGR

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